BRAC UNIVERSITY

Department of Computer Science and Engineering

CSE 461: Introduction to Robotics

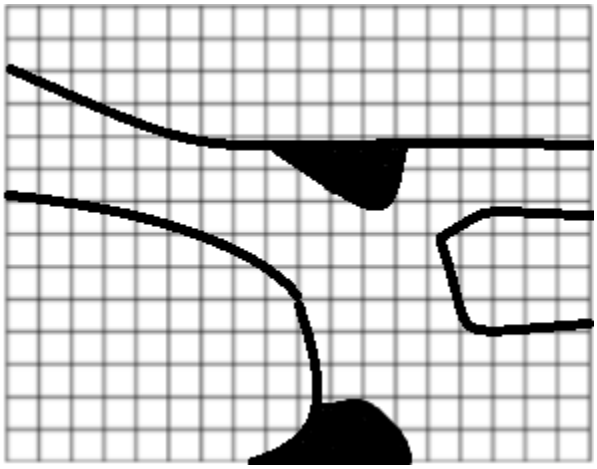
Examination: Quiz IV **Duration: 25min**
Semester : Summer 2026 **Full Marks: 15**

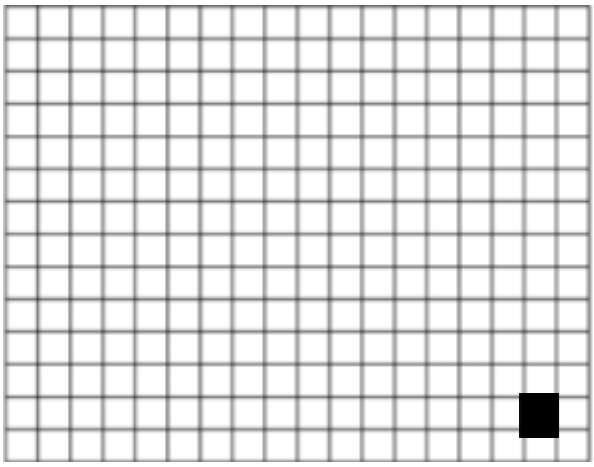
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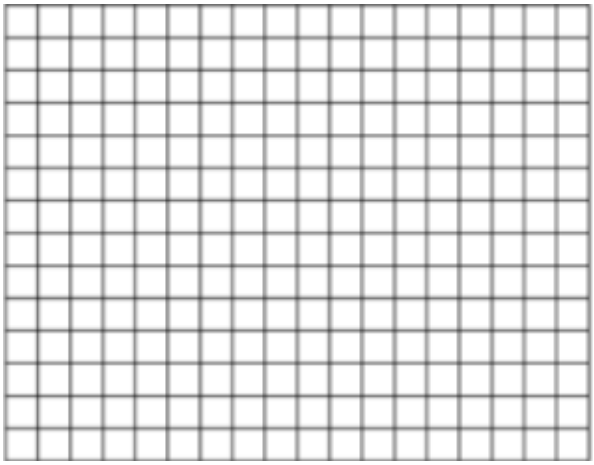
1	**In an unknown environment, which statement is MOST accurate?** A. Path planning must always be global B. Exploration guarantees optimal path C. Mapping and exploration must occur simultaneously D. Localization is unnecessary **Answer: C**	**1**
2	**Which of the following BEST describes a frontier cell?** A. A known occupied cell near obstacles B. A boundary between two occupied regions C. An unknown cell adjacent to at least one empty cell D. A cell farthest from the robot **Answer: C**	**1**
3	**In the depth sensor model, which region corresponds to maximum uncertainty?** A. Region 1 (dist < r) B. Region 2 (dist = r) C. Region 3 (dist > r) D. All regions equally **Answer: C**	**1**
4	**In what scenario would visual homing fail completely, and why?** **Answer:** When the robot gets stuck, can't see the goal, or randomly keeps moving in a loop.	**2**
5	A lunar expedition successfully lands its first rover on the Moon to explore a previously **unmapped** region near the south pole. During the mission, a sudden radiation storm disrupts the rover's memory, unable to store new info and corrupts the file that had the co-ordinates of the base. But the file that stored the base's image was still untouched along with its world class sensors. As the chief navigation officer at mission control, you must guide the rover back to its known base station to recharge before its power runs out.	**10**

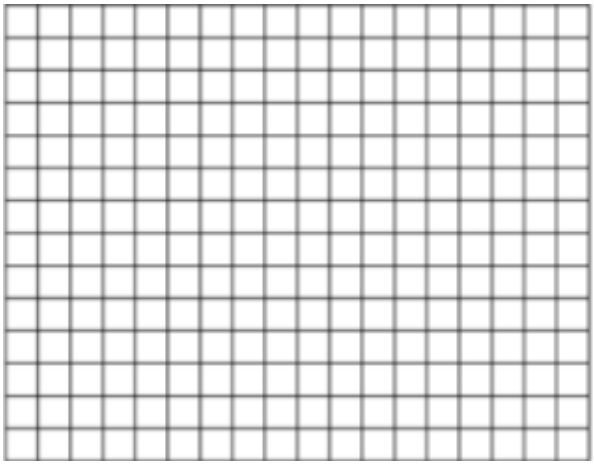
A. **Which** path planning method would you choose? Defend your answer based on the scenario.**[2]**
B. Which localization method would be most suitable and why? **[2]**
C. **Simulate** the rover's 3 steps across the surface of the moon. The initial position of the robot in the grid is as below in the first figure. **[6]**

Must show all the empty, occupied, unknown and frontier cells

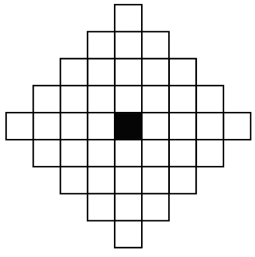








Sensor Model:



Answers:

A. Visual homing. Since the base's co-ordinates are lost, the robot has no idea about it's locations. So, it can not use bug-based approach. Instead, it will randomly move around looking for the base that is similar to the saved image. Once the robot finds the base, it will move directly towards it.

B. Landmark sensing. Since the memory is corrupted, it can not save anything new. Dead-reckoning requires the robot to save every movement it takes. But that is not possible now. Instead the robot can use its sensors to check for the landmarks and try to localize itself.
   State-estimation is also correct.

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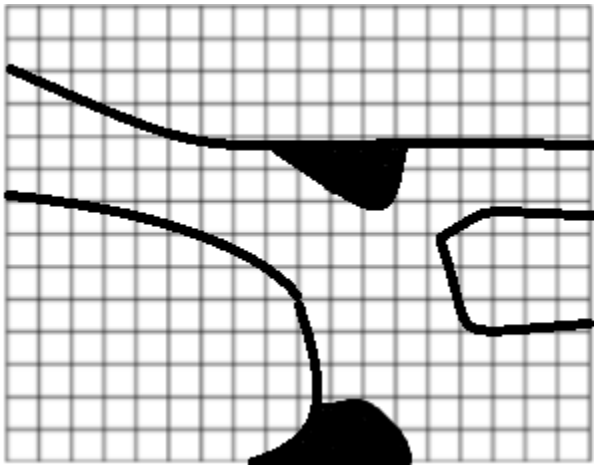
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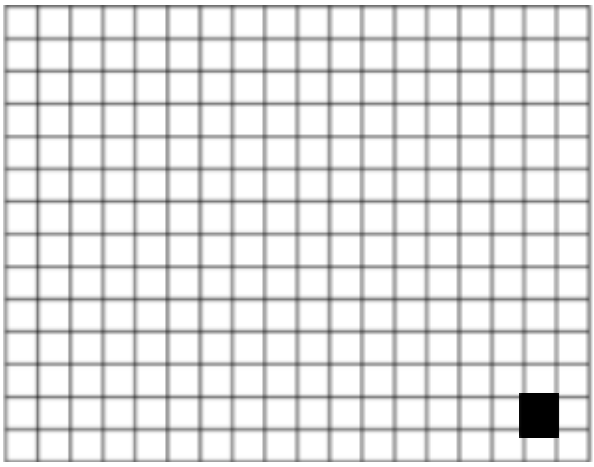
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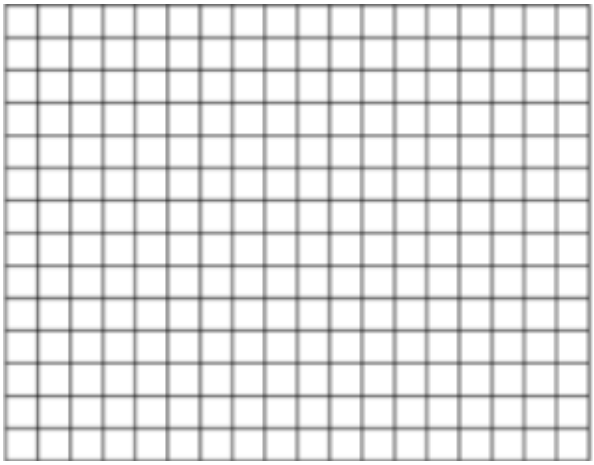
1	**In the depth sensor model, which region corresponds to maximum uncertainty?** A. Region 1 (dist < r) B. Region 2 (dist = r) C. Region 3 (dist > r) D. All regions equally **Answer: C**	**1**
2	**Which approach works without a map but with global goal direction?** A. A* Path Planning B. Visual Homing C. Bug-based algorithms D. Frontier exploration **Answer: C**	**1**
3	**In an unknown environment, which statement is MOST accurate?** A. Path planning must always be global B. Exploration guarantees optimal path C. Mapping and exploration must occur simultaneously D. Localization is unnecessary **Answer: C**	**1**
4	**Why does frontier-based exploration outperform random movement strategies**? **Answer:** Random movement never ensures that the robot will explore the whole map and can keep running in a loop. But frontier cells guarantee that. As long as there are frontier cells in the map, the robot will keep exploring instead of randomly moving around.	**2**
5	A lunar expedition successfully lands its first rover on the Moon to explore a previously **unmapped** region near the south pole. During the mission, a sudden dust storm disrupts visibility and damages the rover's primary navigation system. But the file that contained the base's co-ordinates was still intact. As the chief navigation officer at mission control, you must guide the rover back to its known base station to recharge before its power runs out.	**10**

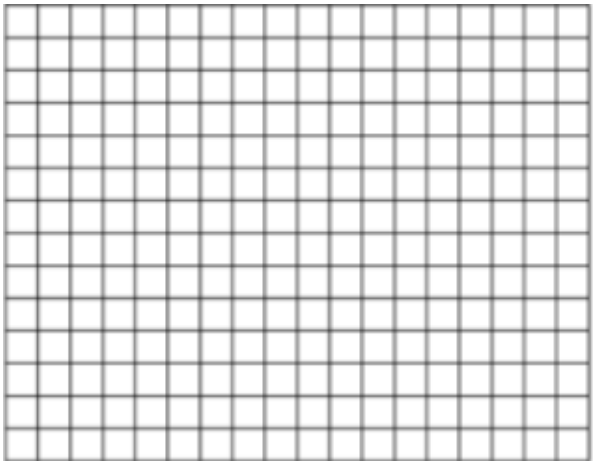
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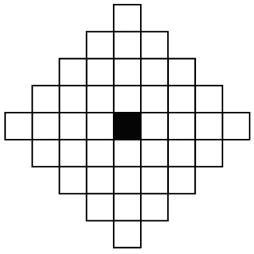








Sensor Model:



Answers:

A. Bug-based Approach. Since visibility is disrupted, the robot can not use its camera to look for the base. So, it can not use visual homing approach. Instead, it will use the base's co-ordinates as its goal to slowly move towards it while avoiding obstacles along the way like a bug.

B. Dead-reckoning. Since the robot is unable to see around, it can not use landmark sensing. Instead it will use dead-reckoning to save every movement it takes. And gradually find its way back to the base.
   State-estimation is also correct.